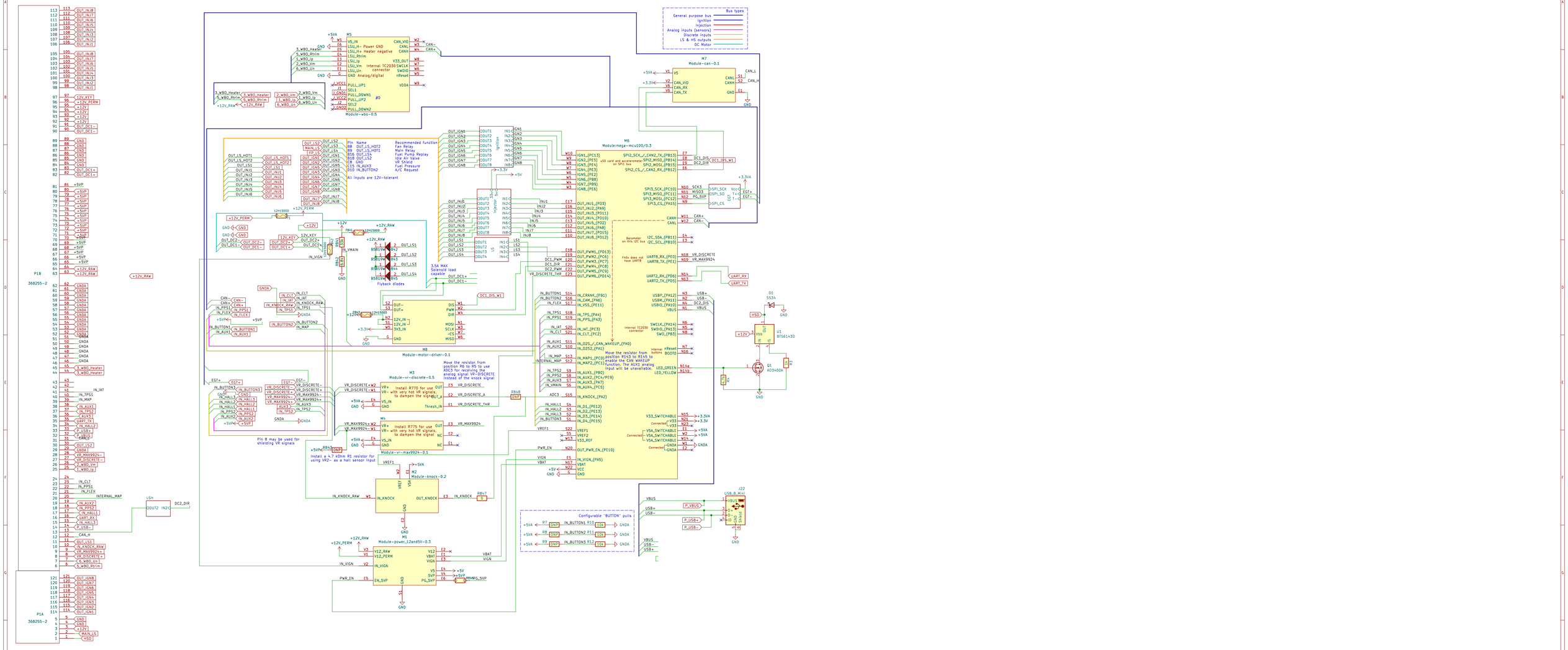
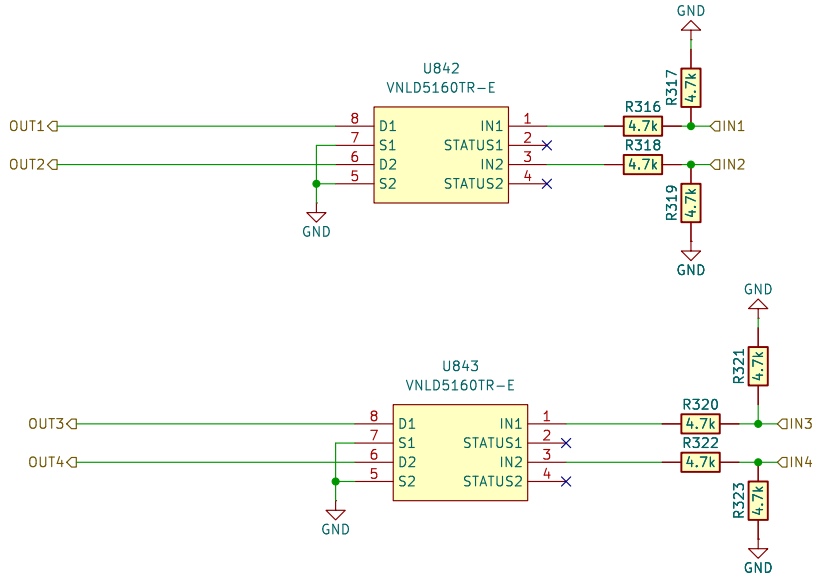
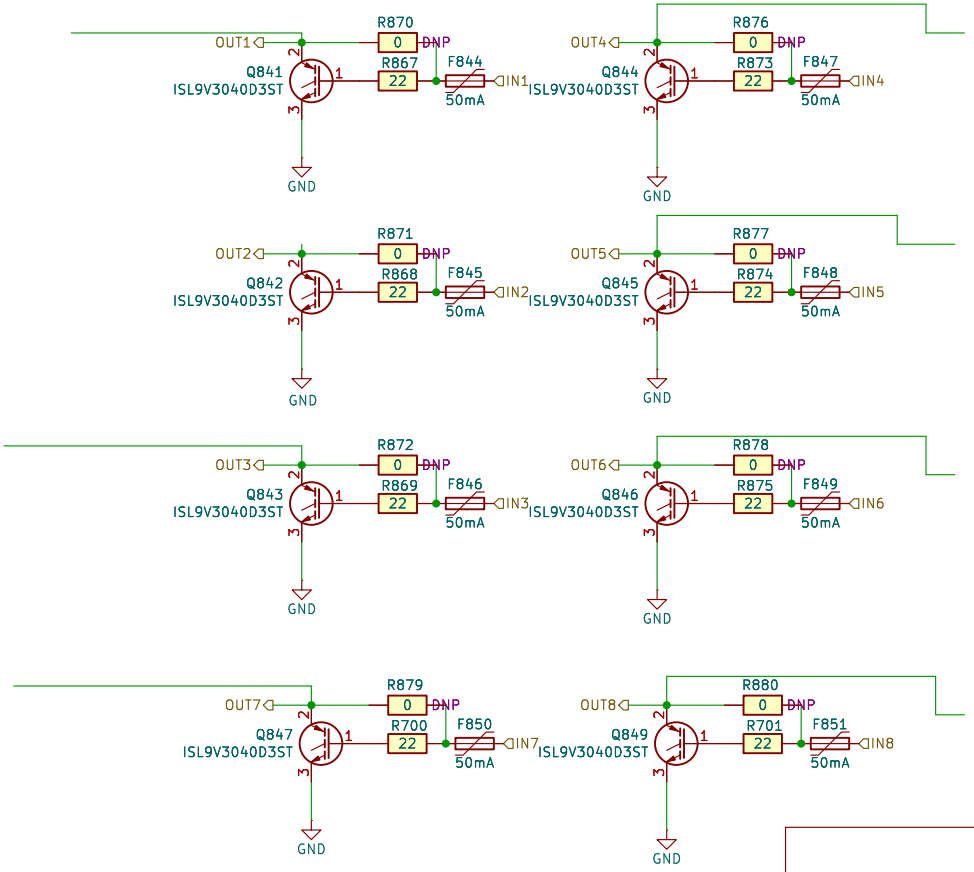

Bus types
General purpose bus
Ignition
Injection
Analog inputs (sensors)
Discrete inputs
LS & HS outputs
DC Motor
M5
Module-wbo-0.5
M7
Module-can-0.1
M6
Module mega-mcu100/0.3
SD card and accessories on SPI1 bus
Barometer on this I2C bus
F4Dv does not have UART8
Internal TC2030 connector
Internal buttons
Move the resistor from position R6 to P5 to use ADC5 for receiving the IN_DISCRETE_THR signal instead of the knock signal
M3
Module-vr-discrete-0.5
Install R770 for use with very hot VR signals, to dampen the signal
M8
Module-motor-driver-0.1
M4
Module-vr-max9924-0.3
Install R775 for use with very hot VR signals, to dampen the signal
Pin 8 may be used for shielding VR signals
Install a 4.7 kOhm R1 resistor for using VR2- as a hall sensor input
M2
Module-knock-0.2
M1
Module-power_12and5V-0.3
3.5A MAX Solenoid load capable
Flyback diodes
Move the resistor from position R143 to R145 to enable the CAN WAKEUP function. The AUX1 analog input will be unavailable.
Configurable "BUTTON" pulls
P1B
368255-2
P1A
368255-2
LSH
DC2_DIR
J22
USB_B_Mini
U1
BTS61430
Q1
AO3400A
D1
SS34
HS0
LS0
P_VBUS
P_USB-
P_USB+
IN_VIGN
IN_KNOCK_RAW
OUT_KNOCK
IN_KNOCK
VBAT
VIGN
PWR_EN
EN_5VP
PG_5VP
+12V_RAW
+12V_PERM
+5VP
+3.3V
+5VA
+3.3VA
GNDA
GND
Title: Hellen121vag
Sheet: /
File: greenenergyecu.kicad_sch
Size: A1
Date: 2022-10-14
KiCad E.D.A. 8.0.9
Rev: d
Id: 1/6
andreika ● rusEFI.comm

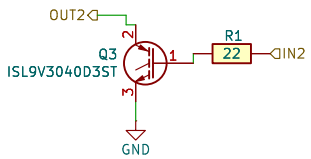


Sheet: /LS/		
File: LS4.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.9		Id: 3/6

Remove R20–R22, R26–R28, and install Q7–Q12 for use with dumb coils



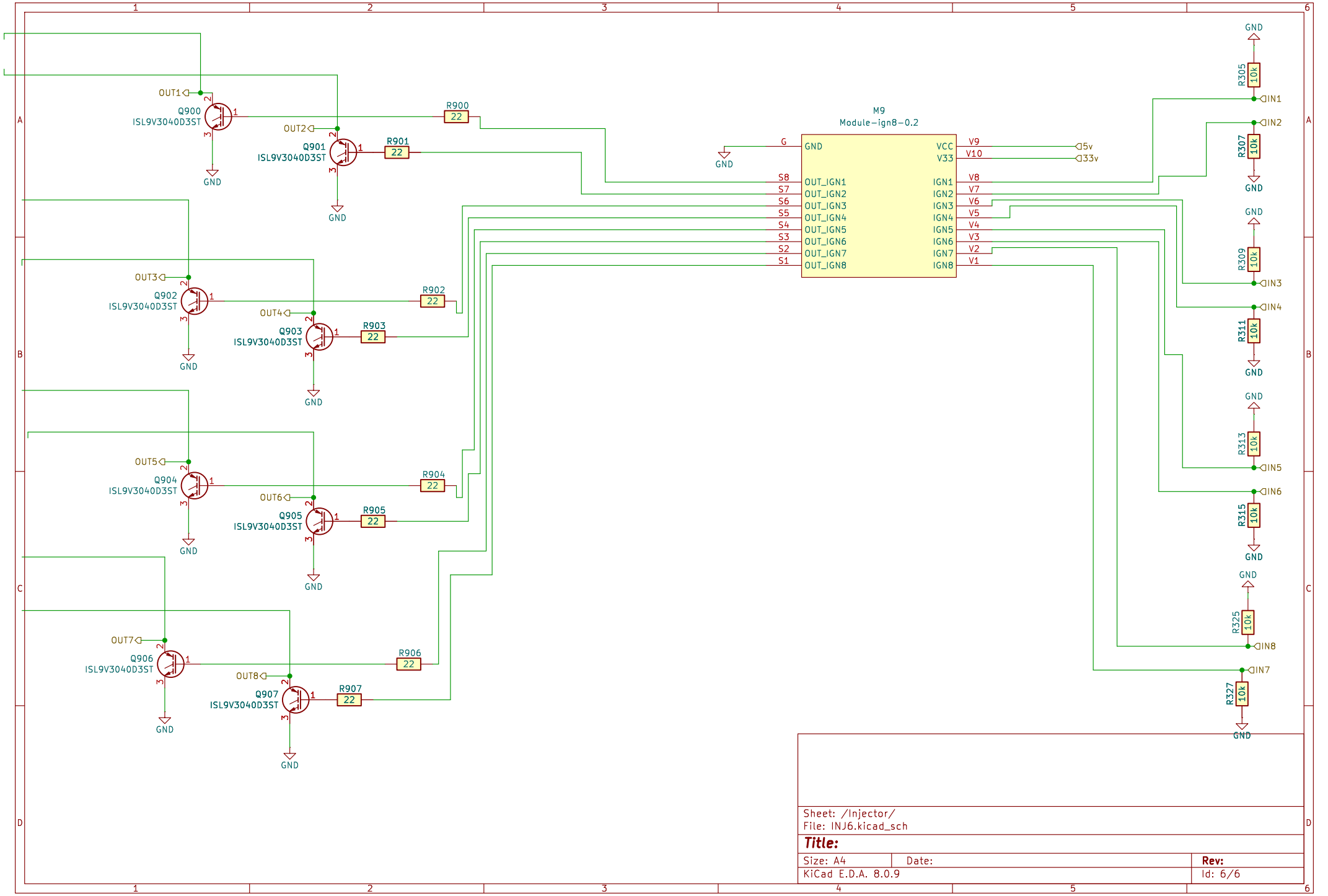
Sheet: /Ignition/		
File: IGN6.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.9		Id: 5/6

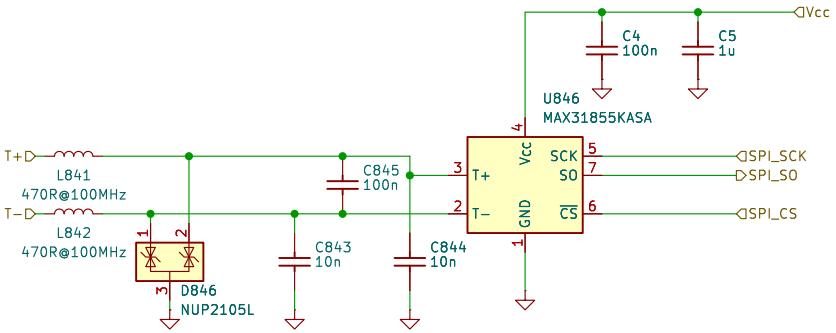


Sheet: /LSH/
File: LS_H.kicad_sch

Title:

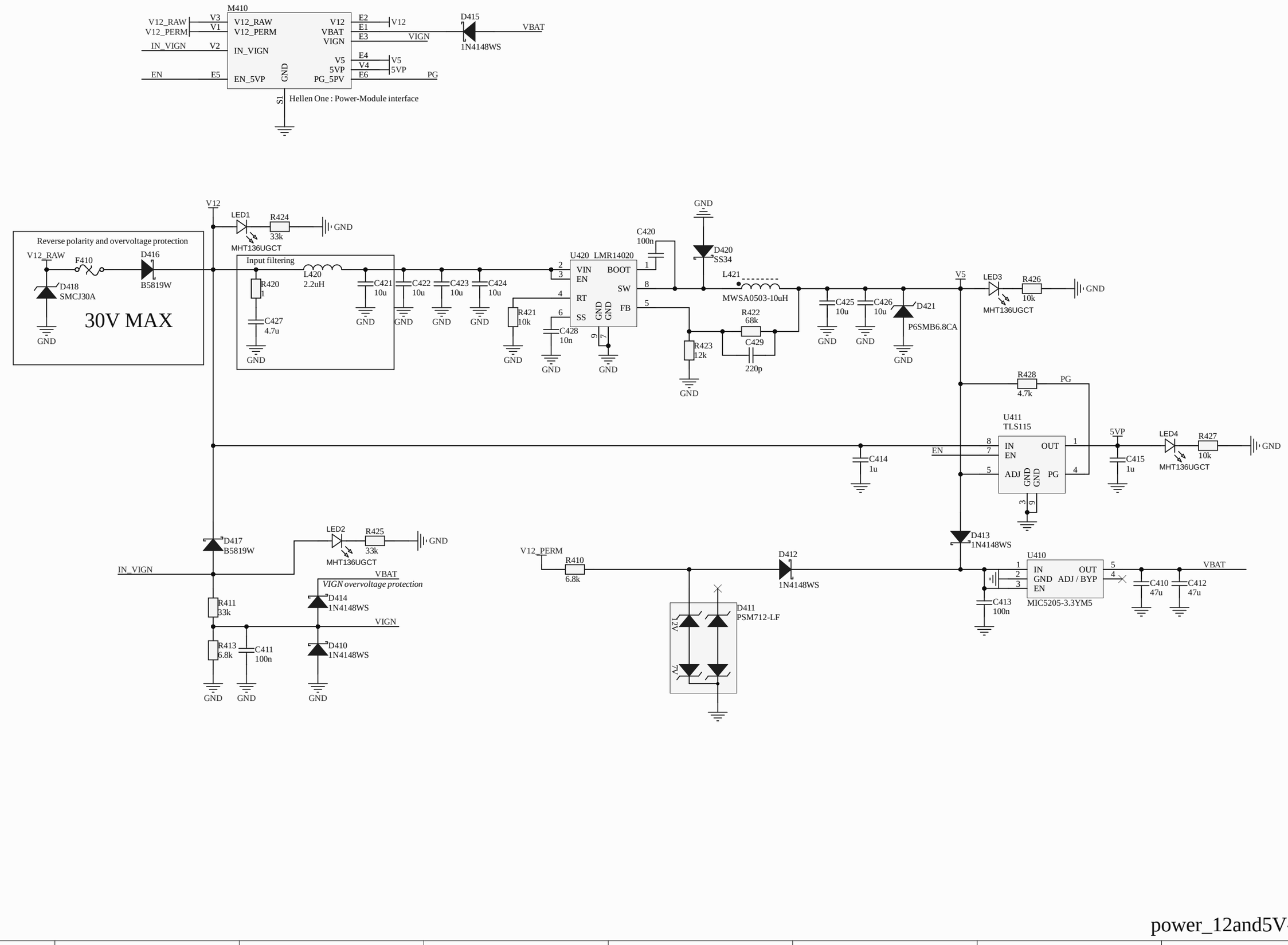
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.9		Id: 6/6


M9
Module-ign8-0.2
G
GND
S8
S7
S6
S5
S4
S3
S2
S1
OUT_IGN1
OUT_IGN2
OUT_IGN3
OUT_IGN4
OUT_IGN5
OUT_IGN6
OUT_IGN7
OUT_IGN8
VCC
V33
V9
V10
V8
V7
V6
V5
V4
V3
V2
V1
IGN1
IGN2
IGN3
IGN4
IGN5
IGN6
IGN7
IGN8
5v
33v
OUT1
OUT2
OUT3
OUT4
OUT5
OUT6
OUT7
OUT8
Q900
Q901
Q902
Q903
Q904
Q905
Q906
Q907
ISL9V3040D3ST
R900
R901
R902
R903
R904
R905
R906
R907
22
R305
R307
R309
R311
R313
R315
R325
R327
10k
IN1
IN2
IN3
IN4
IN5
IN6
IN7
IN8
Sheet: /Injector/
File: INJ6.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 8.0.9
Id: 6/6

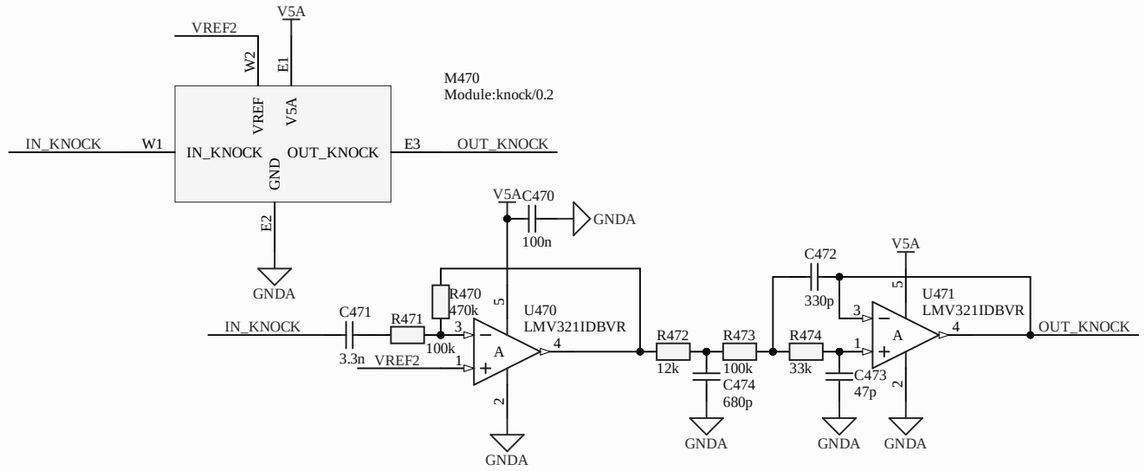


NUP2105L is optional, used for protection

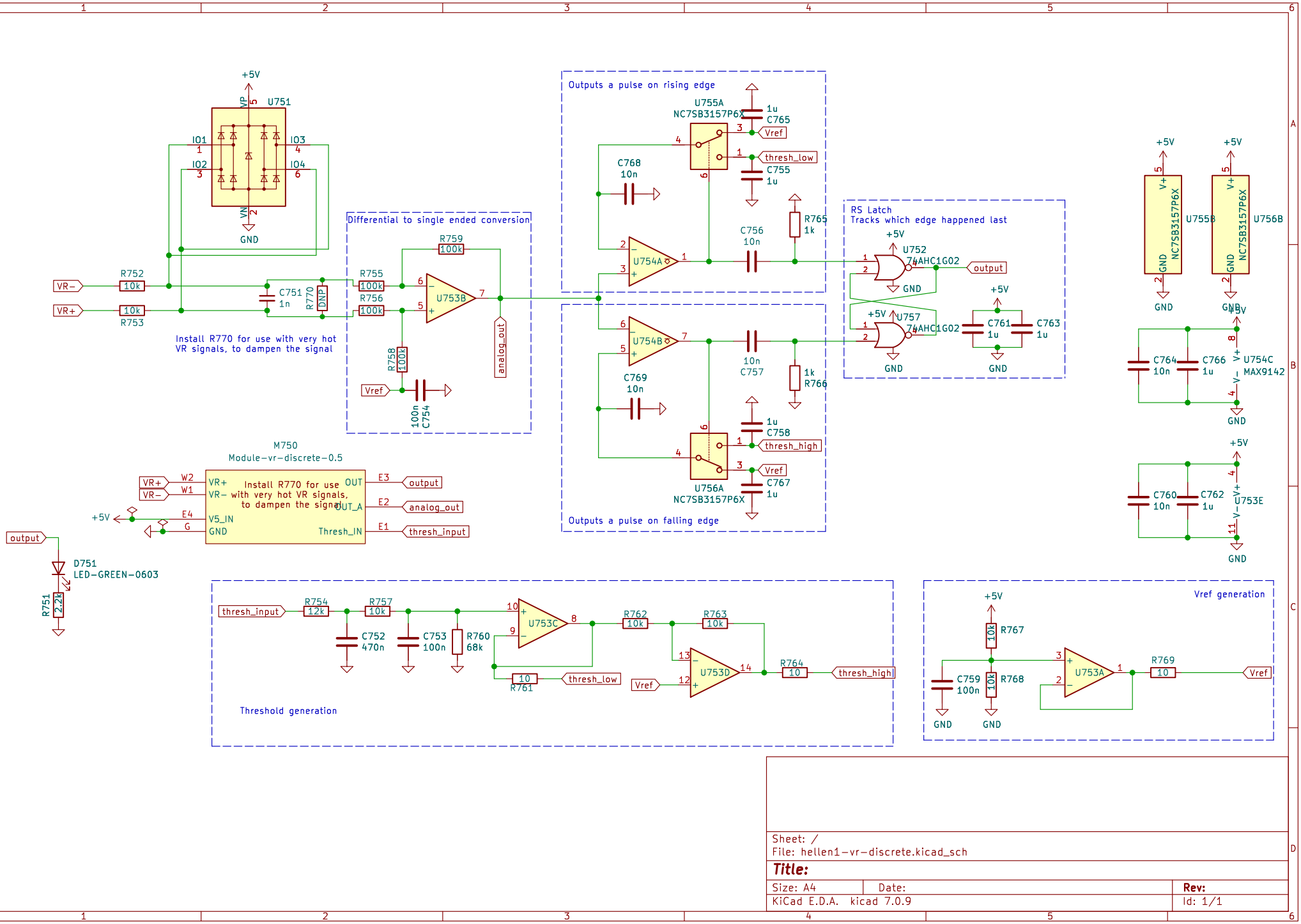
Sheet: /EGT/		
File: egt.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.9		Id: 9/6



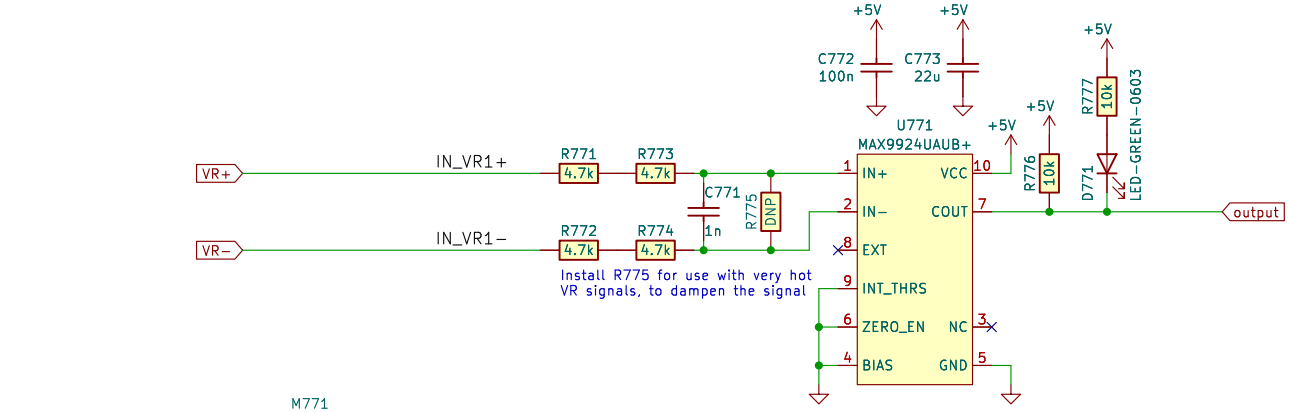
power_12and5V-0.3



Title: ***Hellen-One Knock***			***rusEFI EMS*** ***Hellen-One Platform*** *https://rusefi.com/s/hellen72*
Size: A3	Author: andreika	Revision: 0.2	
Date: 15.02.2023		Sheet 1 of 1	



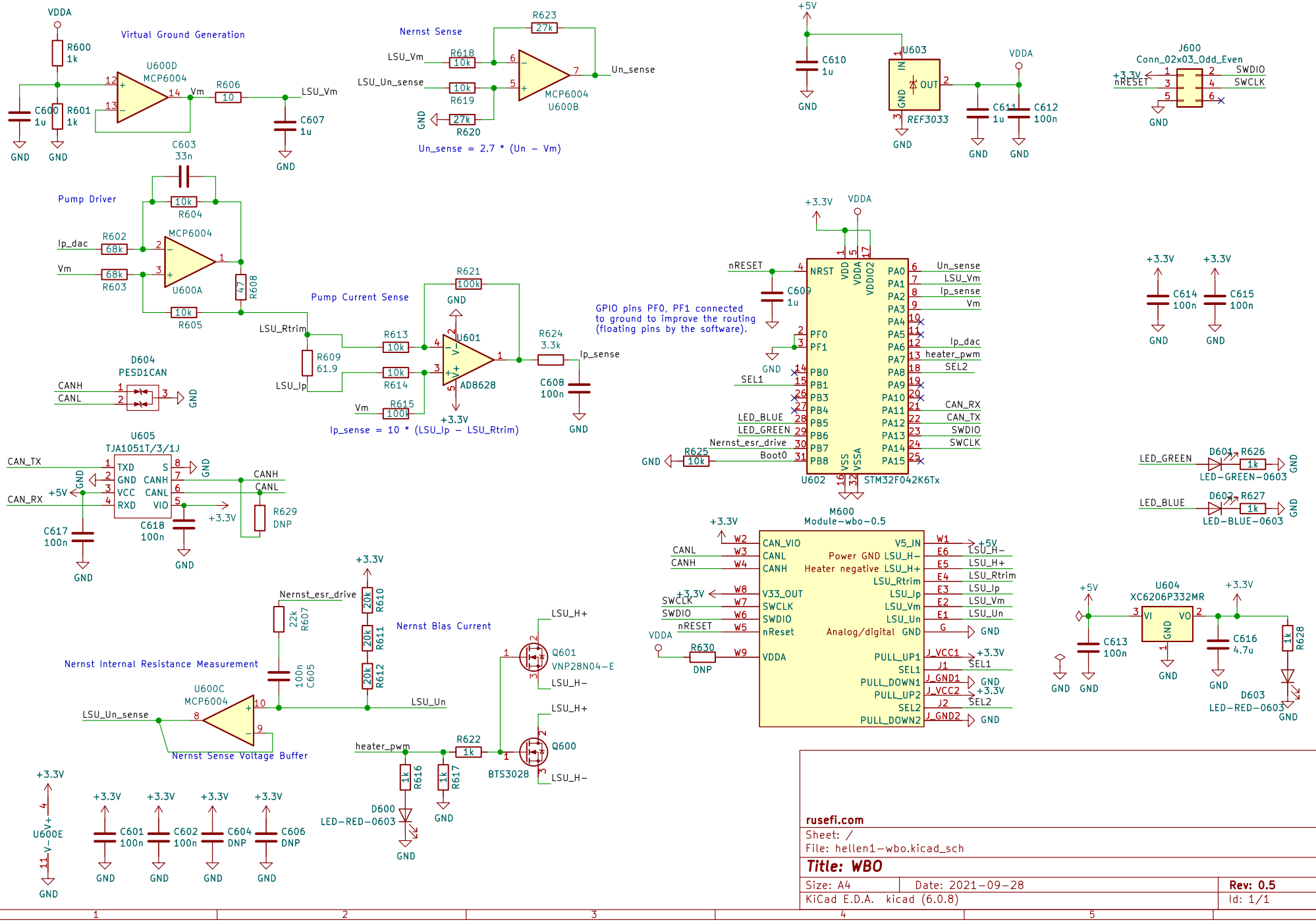
Sheet: /		
File: hellen1−vr−discrete.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. kicad 7.0.9		Id: 1/1

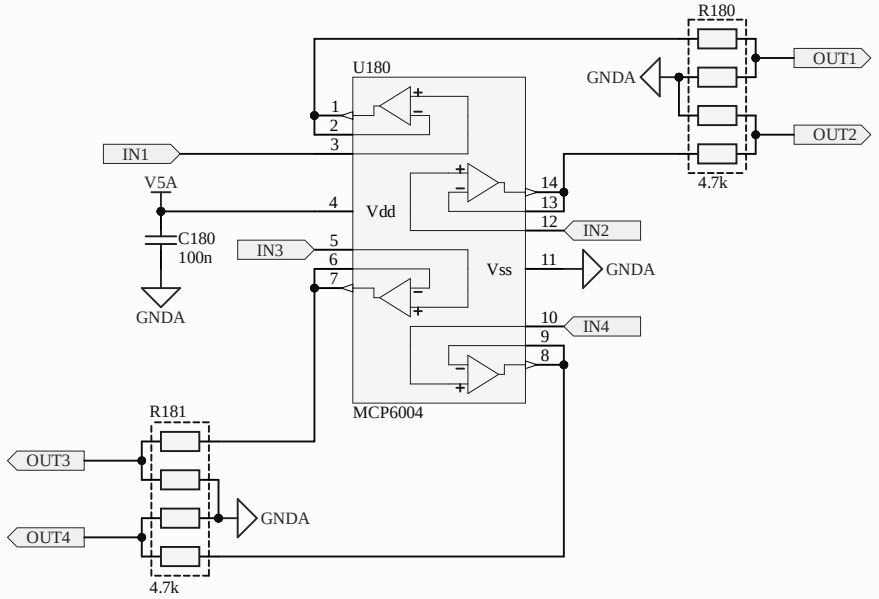


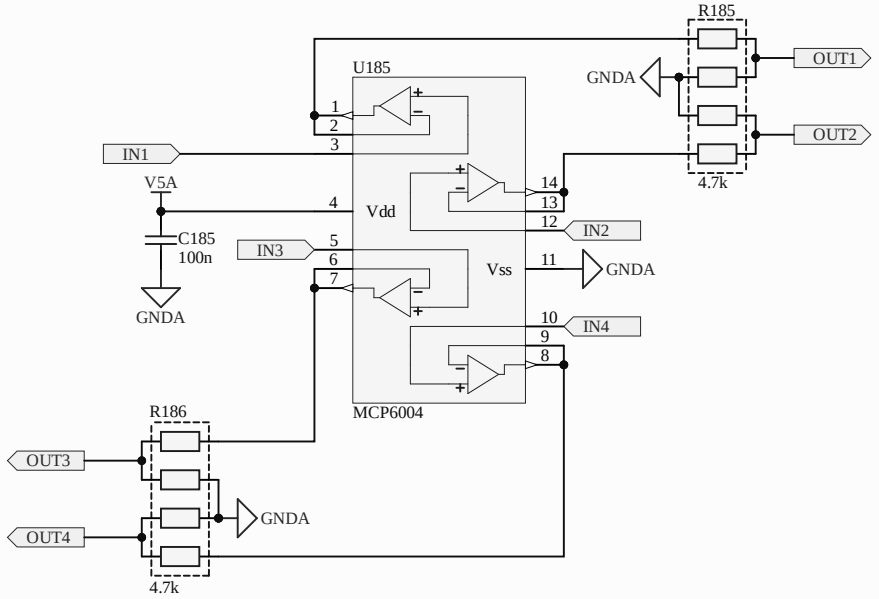
Sheet: /
File: hellen1−vr−max9924.kicad_sch

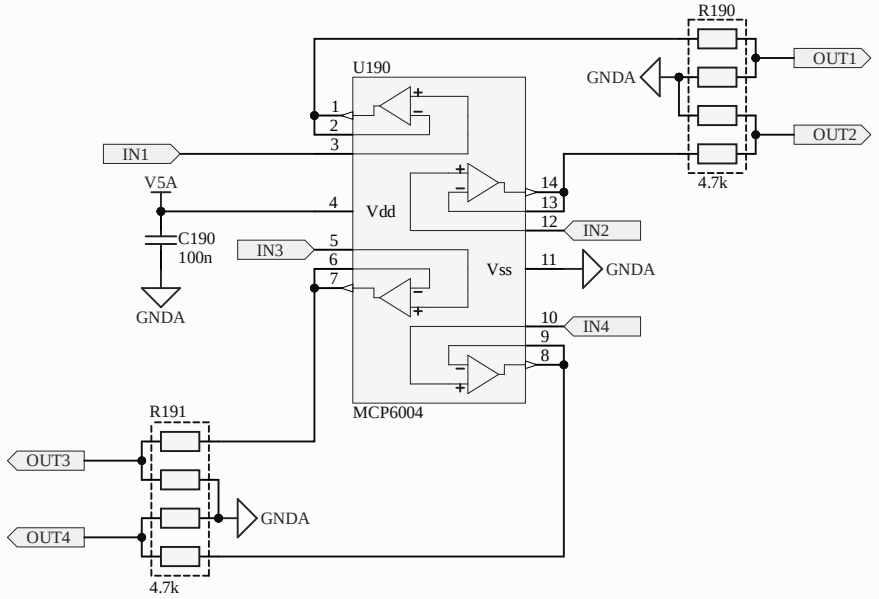
Title: VR MAX9924 module

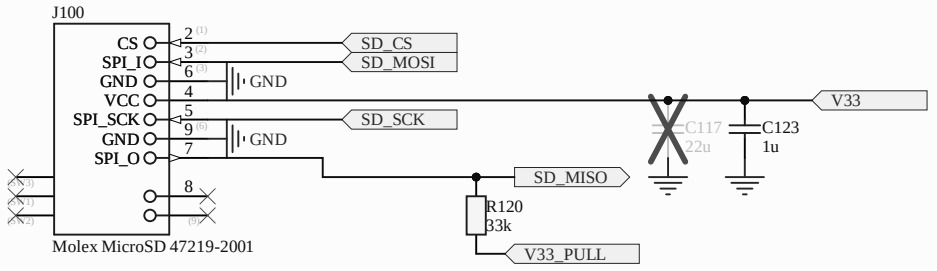
Size: A4	Date: 2023−07−25	**Rev: 0.1**
KiCad E.D.A. eeschema 7.0.9		Id: 1/1

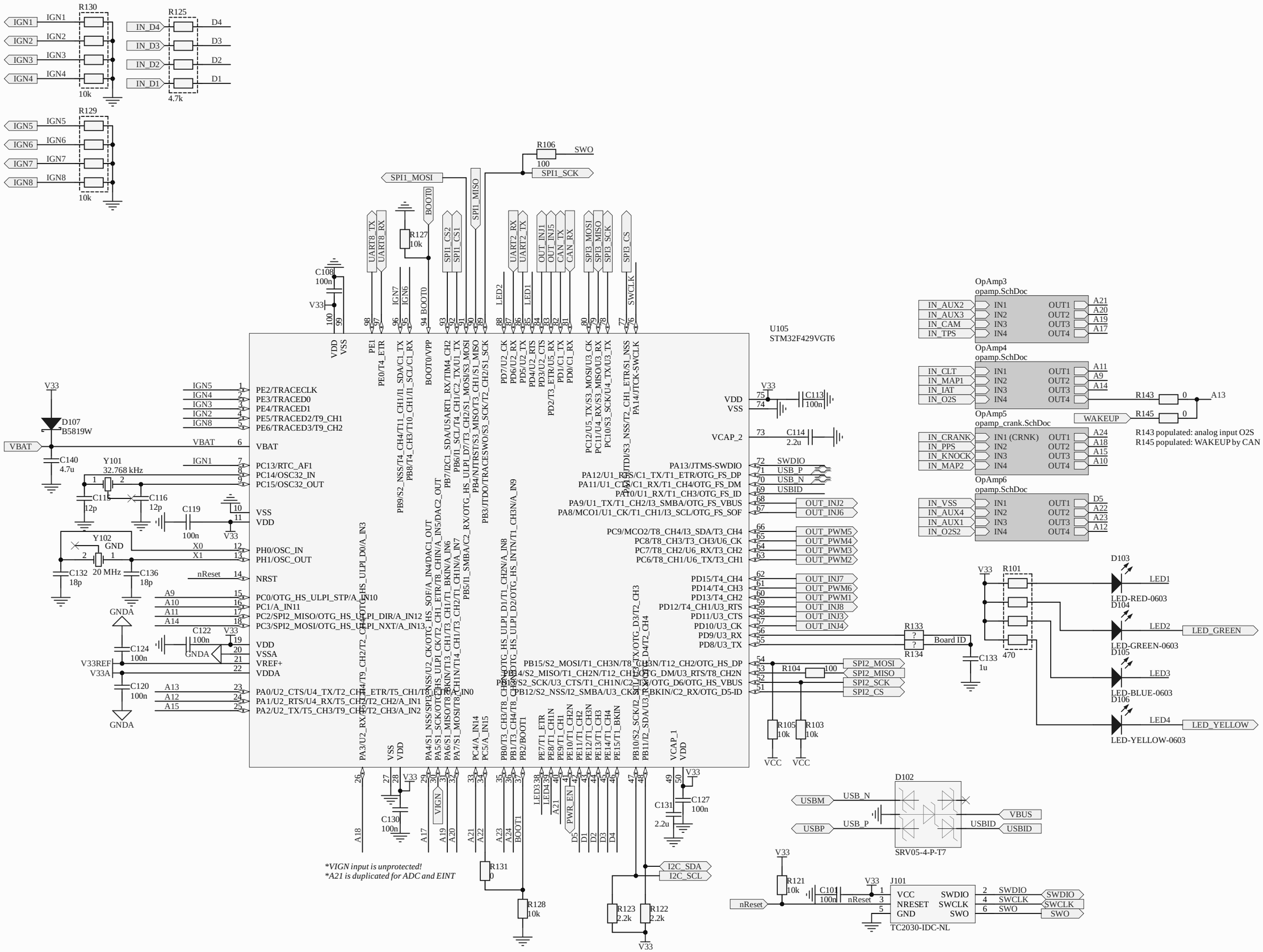

Virtual Ground Generation
U600D
MCP6004
R600 1k
R601 1k
C600 1u
R606 10
C607 1u
LSU_Vm
Vm
Nernst Sense
R623 27k
R618 10k
R619 10k
R620 27k
LSU_Vm
LSU_Un_sense
Un_sense
U600B
Un_sense = 2.7 * (Un − Vm)
+5V
C610 1u
U603
REF3033
C611 1u
C612 100n
J600
Conn_02x03_Odd_Even
SWDIO
SWCLK
nRESET
Pump Driver
C603 33n
R604 10k
R605 10k
R602 68k
R603 68k
R608 47
U600A
Ip_dac
Vm
LSU_Rtrim
LSU_Ip
Pump Current Sense
R621 100k
R613 10k
R614 10k
R615 100
R609 61.9
U601
AD8628
R624 3.3k
Ip_sense
C608 100n
Ip_sense = 10 * (LSU_Ip − LSU_Rtrim)
D604
PESD1CAN
CANH
CANL
U605
TJA1051T/3/1J
TXD
GND
VCC
RXD
CANH
CANL
S
VIO
C617 100n
C618 100n
R629 DNP
GPIO pins PF0, PF1 connected
to ground to improve the routing
(floating pins by the software).
U602
STM32F042K6Tx
nRESET
C609 1u
SEL1
LED_BLUE
LED_GREEN
Nernst_esr_drive
R625 10k
Boot0
Un_sense
LSU_Vm
Ip_sense
Vm
Ip_dac
heater_pwm
SEL2
CAN_RX
CAN_TX
SWDIO
SWCLK
C614 100n
C615 100n
M600
Module-wbo-0.5
CAN_VIO
CANL
CANH
V33_OUT
SWCLK
SWDIO
nReset
VDDA
R630 DNP
V5_IN
Power GND LSU_H−
Heater negative LSU_H+
LSU_Rtrim
LSU_Ip
LSU_Vm
LSU_Un
Analog/digital GND
PULL_UP1
SEL1
PULL_DOWN1
PULL_UP2
SEL2
PULL_DOWN2
LSU_H−
LSU_H+
LSU_Un
LED_GREEN
D601
R626 1k
LED−GREEN−0603
LED_BLUE
D602
R627 1k
LED−BLUE−0603
U604
XC6206P332MR
C613 100n
C616 4.7u
R628 1k
D603
LED−RED−0603
Nernst_esr_drive
R607 22k
R610 20k
R611 20k
R612 20k
C605 100n
Nernst Bias Current
Nernst Internal Resistance Measurement
U600C
MCP6004
LSU_Un_sense
Nernst Sense Voltage Buffer
Q601
VNP28N04−E
Q600
BTS3028
heater_pwm
R622 1k
R616 1k
R617 1k
D600
LED−RED−0603
U600E
C601 100n
C602 100n
C604 DNP
C606 DNP
rusefi.com
Sheet: /
File: hellen1−wbo.kicad_sch
Title: WBO
Size: A4
Date: 2021−09−28
Rev: 0.5
KiCad E.D.A. kicad (6.0.8)
Id: 1/1


R180
U180
GNDA
OUT1
OUT2
IN1
V5A
Vdd
4.7k
C180
100n
IN3
IN2
Vss
GNDA
GNDA
IN4
MCP6004
R181
OUT3
GNDA
OUT4
4.7k

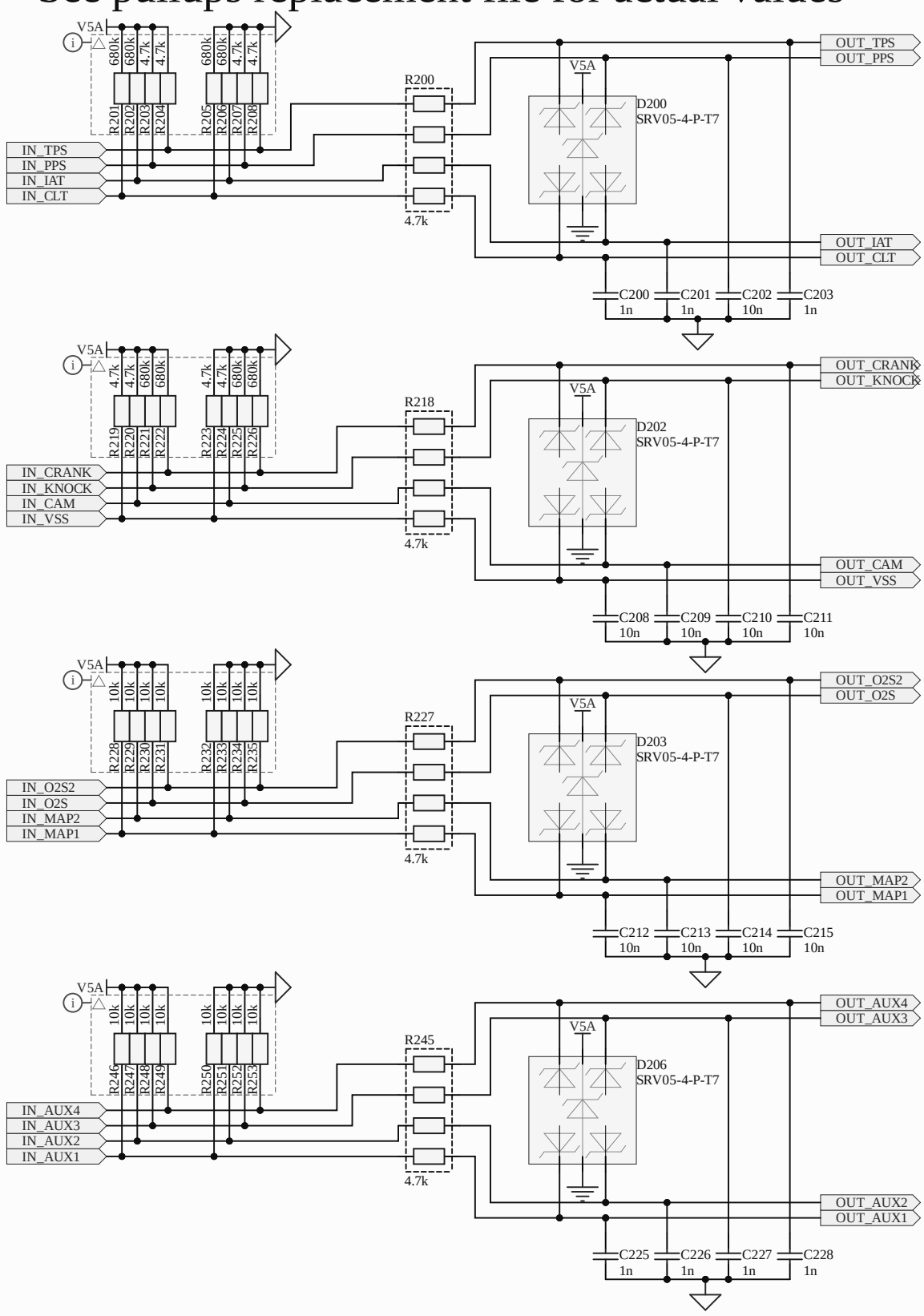

R185
U185
GNDA
OUT1
OUT2
IN1
V5A
Vdd
4.7k
C185
100n
IN3
IN2
Vss
GNDA
GNDA
IN4
MCP6004
R186
OUT3
GNDA
OUT4
4.7k

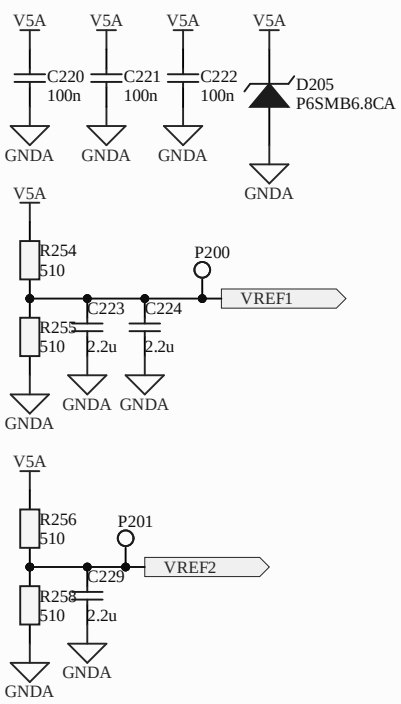

R190
U190
GNDA
OUT1
OUT2
IN1
V5A
Vdd
4.7k
C190
100n
IN3
IN2
Vss
GNDA
GNDA
IN4
MCP6004
R191
OUT3
GNDA
OUT4
4.7k

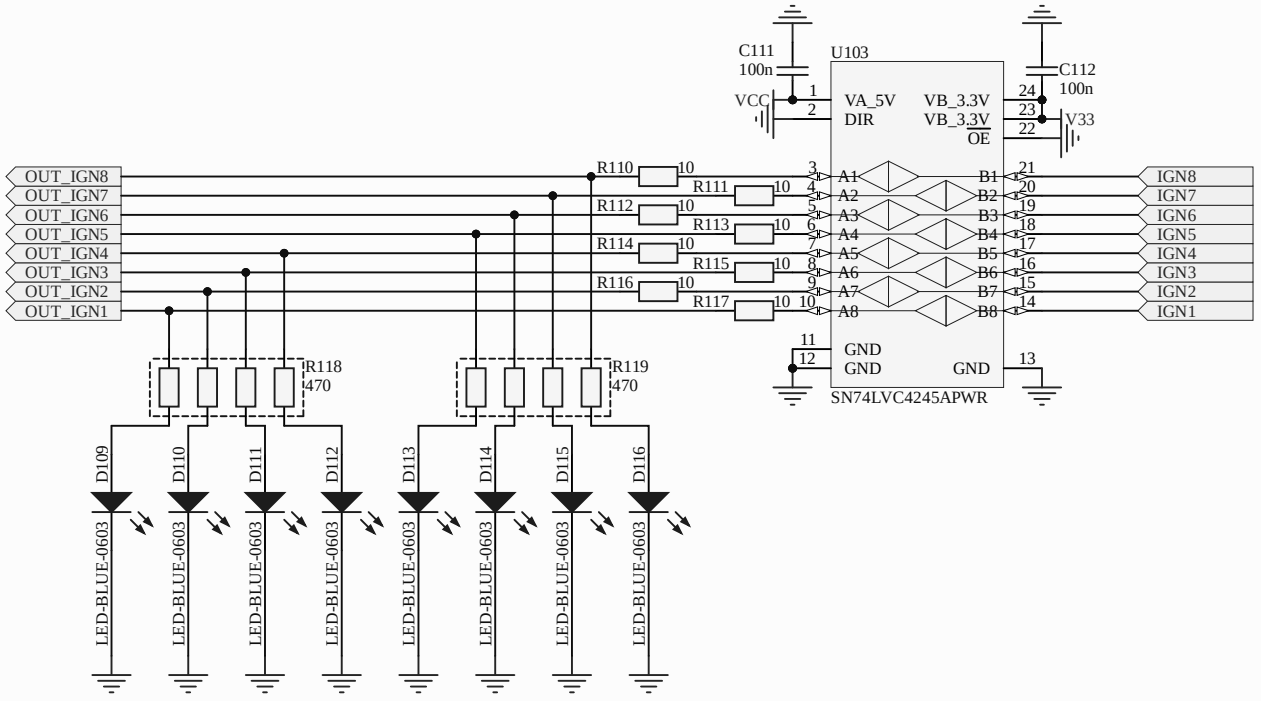

J100
CS
SPI_I
GND
VCC
SPI_SCK
GND
SPI_O
2
3
6
4
5
9
7
8
SD_CS
SD_MOSI
GND
SD_SCK
GND
SD_MISO
R120
33k
V33_PULL
C117
22u
C123
1u
V33
Molex MicroSD 47219-2001


R130
IGN1
IGN2
IGN3
IGN4
10k
R125
IN_D4
IN_D3
IN_D2
IN_D1
D4
D3
D2
D1
4.7k
R129
IGN5
IGN6
IGN7
IGN8
10k
R106
SWO
100
SPI1_SCK
SPI1_MOSI
SPI1_MISO
BOOT0
R127
10k
C108
100n
V33
UART8_TX
UART8_RX
SPI1_CS2
SPI1_CS1
UART2_RX
UART2_TX
OUT_INJ1
OUT_INJ5
CAN_TX
CAN_RX
SPI3_MOSI
SPI3_MISO
SPI3_SCK
SPI3_CS
SWCLK
LED2
LED1
IGN7
IGN6
BOOT0
U105
STM32F429VGT6
VDD
VSS
PE1
PE0/T4_ETR
BOOT0/VPP
PD7/U2_CK
PD6/U2_RX
PD5/U2_TX
PD4/U2_RTS
PD3/U2_CTS
PD2/T3_ETR/U5_RX
PD1/C1_TX
PD0/C1_RX
PC12/U5_TX/S3_MOSI/U3_CK
PC11/U4_RX/S3_MISO/U3_RX
PC10/S3_SCK/U4_TX/U3_TX
PA14/JTCK-SWCLK
PA13/JTMS-SWDIO
PA12/U1_RTS/C1_TX/T1_ETR/OTG_FS_DP
PA11/U1_CTS/C1_RX/T1_CH4/OTG_FS_DM
PA10/U1_RX/T1_CH3/OTG_FS_ID
PA9/U1_TX/T1_CH2/I3_SMBA/OTG_FS_VBUS
PA8/MCO1/U1_CK/T1_CH1/I3_SCL/OTG_FS_SOF
PC9/MCO2/T8_CH4/I3_SDA/T3_CH4
PC8/T8_CH3/T3_CH3/U6_CK
PC7/T8_CH2/U6_RX/T3_CH2
PC6/T8_CH1/U6_TX/T3_CH1
PD15/T4_CH4
PD14/T4_CH3
PD13/T4_CH2
PD12/T4_CH1/U3_RTS
PD11/U3_CTS
PD10/U3_CK
PD9/U3_RX
PD8/U3_TX
PB15/S2_MOSI/T1_CH3N/T8_CH3N/T12_CH2/OTG_HS_DP
PB14/S2_MISO/T1_CH2N/T12_CH1/OTG_HS_DM/U3_RTS/T8_CH2N
PB13/S2_SCK/U3_CTS/T1_CH1N/C2_TX/OTG_D6/OTG_HS_VBUS
PB12/S2_NSS/I2_SMBA/U3_CK/T1_BKIN/C2_RX/OTG_D5-ID
SPI2_MOSI
SPI2_MISO
SPI2_SCK
SPI2_CS
R104
100
R105
10k
R103
10k
VCC
R133
R134
Board ID
C133
1u
R101
470
D103
D104
D105
D106
LED-RED-0603
LED-GREEN-0603
LED-BLUE-0603
LED-YELLOW-0603
LED_GREEN
LED_YELLOW
OpAmp3
opamp.SchDoc
IN_AUX2
IN_AUX3
IN_CAM
IN_TPS
IN1
IN2
IN3
IN4
OUT1
OUT2
OUT3
OUT4
A21
A20
A19
A17
OpAmp4
opamp.SchDoc
IN_CLT
IN_MAP1
IN_IAT
IN_O2S
A11
A9
A14
A13
R143
0
WAKEUP
R145
OpAmp5
opamp_crank.SchDoc
IN_CRANK
IN_PPS
IN_KNOCK
IN_MAP2
IN1 (CRNK)
A24
A18
A15
A10
R143 populated: analog input O2S
R145 populated: WAKEUP by CAN
OpAmp6
opamp.SchDoc
IN_VSS
IN_AUX4
IN_AUX1
IN_O2S2
D5
A22
A23
A12
V33
VBAT
D107
B5819W
C140
4.7u
Y101
32.768 kHz
C115
12p
C116
12p
C119
100n
Y102
GND
20 MHz
C132
18p
C136
18p
GNDA
C124
100n
C122
100n
V33REF
V33A
C120
100n
C113
100n
C114
2.2u
VCAP_2
SWDIO
USB_P
USB_N
USBID
VCAP_1
C127
100n
C131
2.2u
C130
100n
R131
0
R128
10k
R123
2.2k
R122
2.2k
I2C_SDA
I2C_SCL
*VIGN input is unprotected!
*A21 is duplicated for ADC and EINT
D102
USBM
USBP
USB_N
USB_P
VBUS
USBID
SRV05-4-P-T7
R121
10k
C101
100n
nReset
J101
VCC
NRESET
GND
SWDIO
SWCLK
SWO
TC2030-IDC-NL

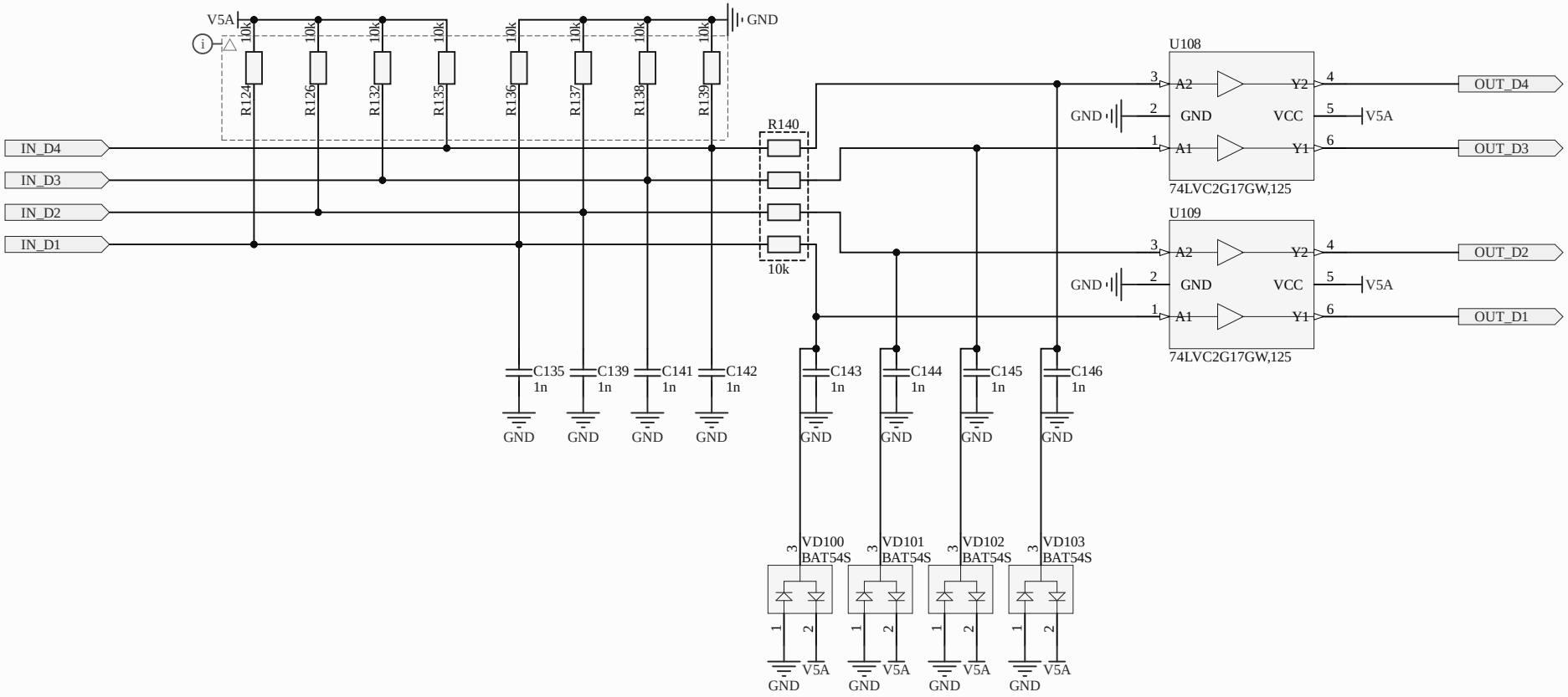
See pullups replacement file for actual values

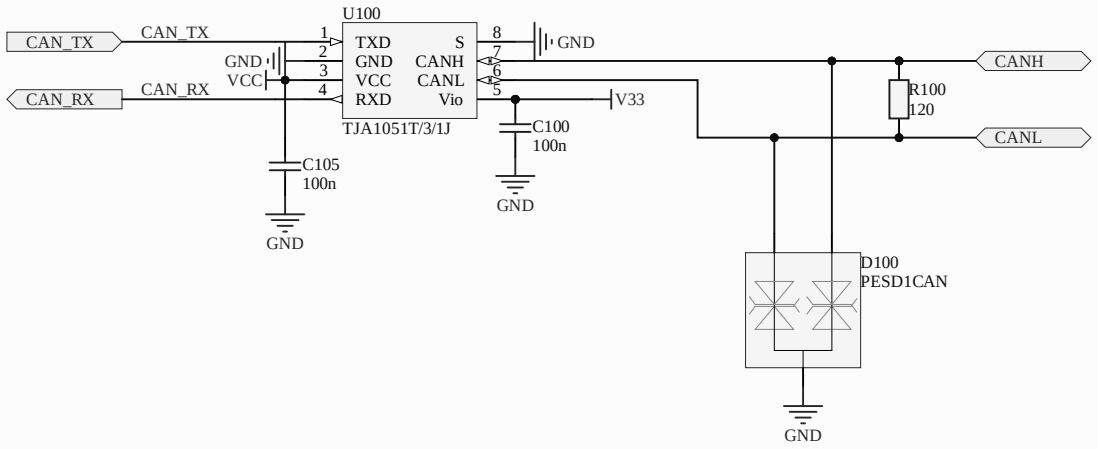


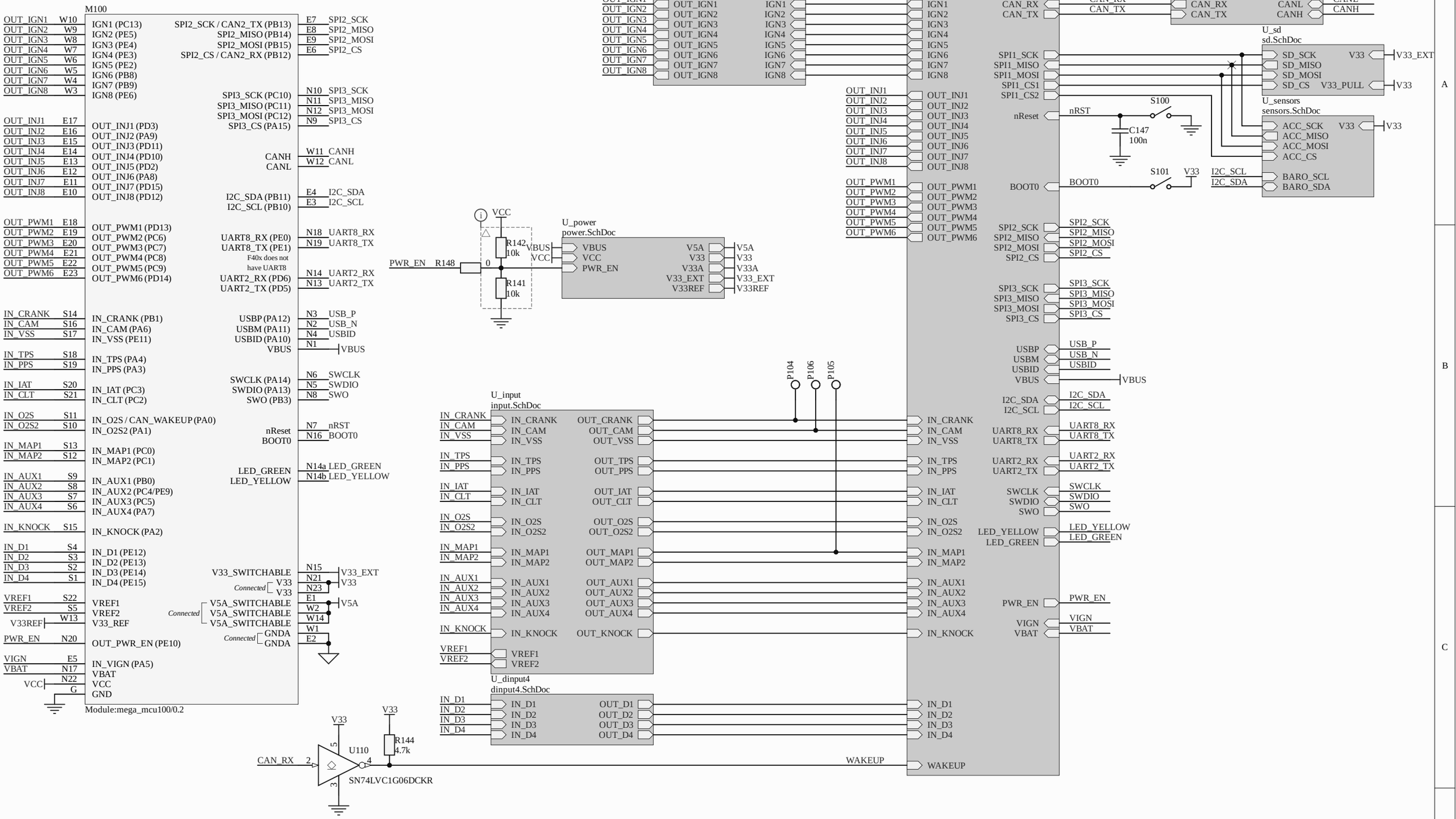


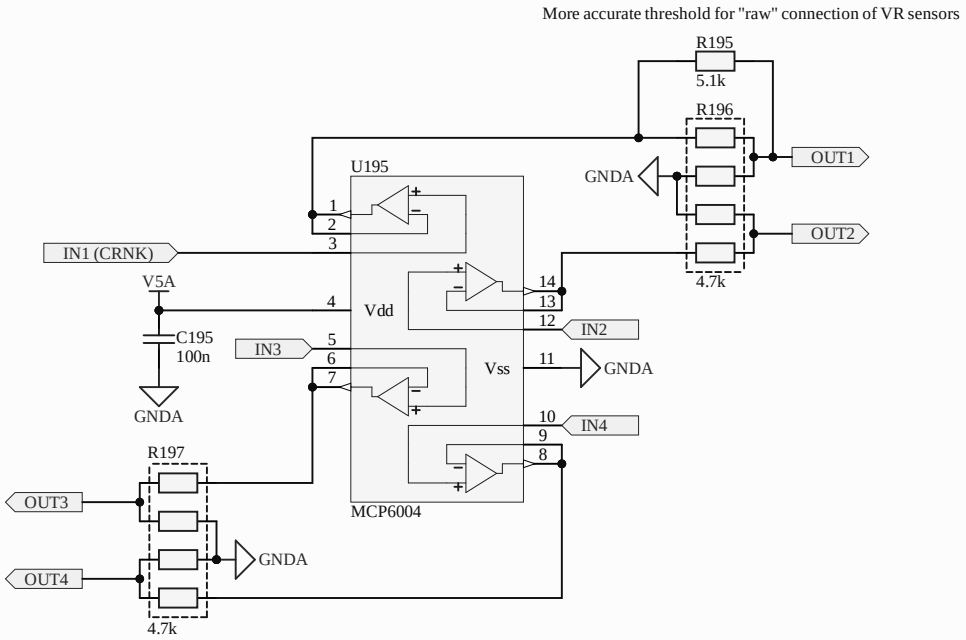

C111
100n
U103
C112
100n
VCC
VA_5V
VB_3.3V
DIR
VB_3.3V
OE
V33
OUT_IGN8
OUT_IGN7
OUT_IGN6
OUT_IGN5
OUT_IGN4
OUT_IGN3
OUT_IGN2
OUT_IGN1
R110
R111
R112
R113
R114
R115
R116
R117
10
A1
A2
A3
A4
A5
A6
A7
A8
B1
B2
B3
B4
B5
B6
B7
B8
IGN8
IGN7
IGN6
IGN5
IGN4
IGN3
IGN2
IGN1
GND
GND
GND
SN74LVC4245APWR
R118
470
R119
470
D109
D110
D111
D112
D113
D114
D115
D116
LED-BLUE-0603

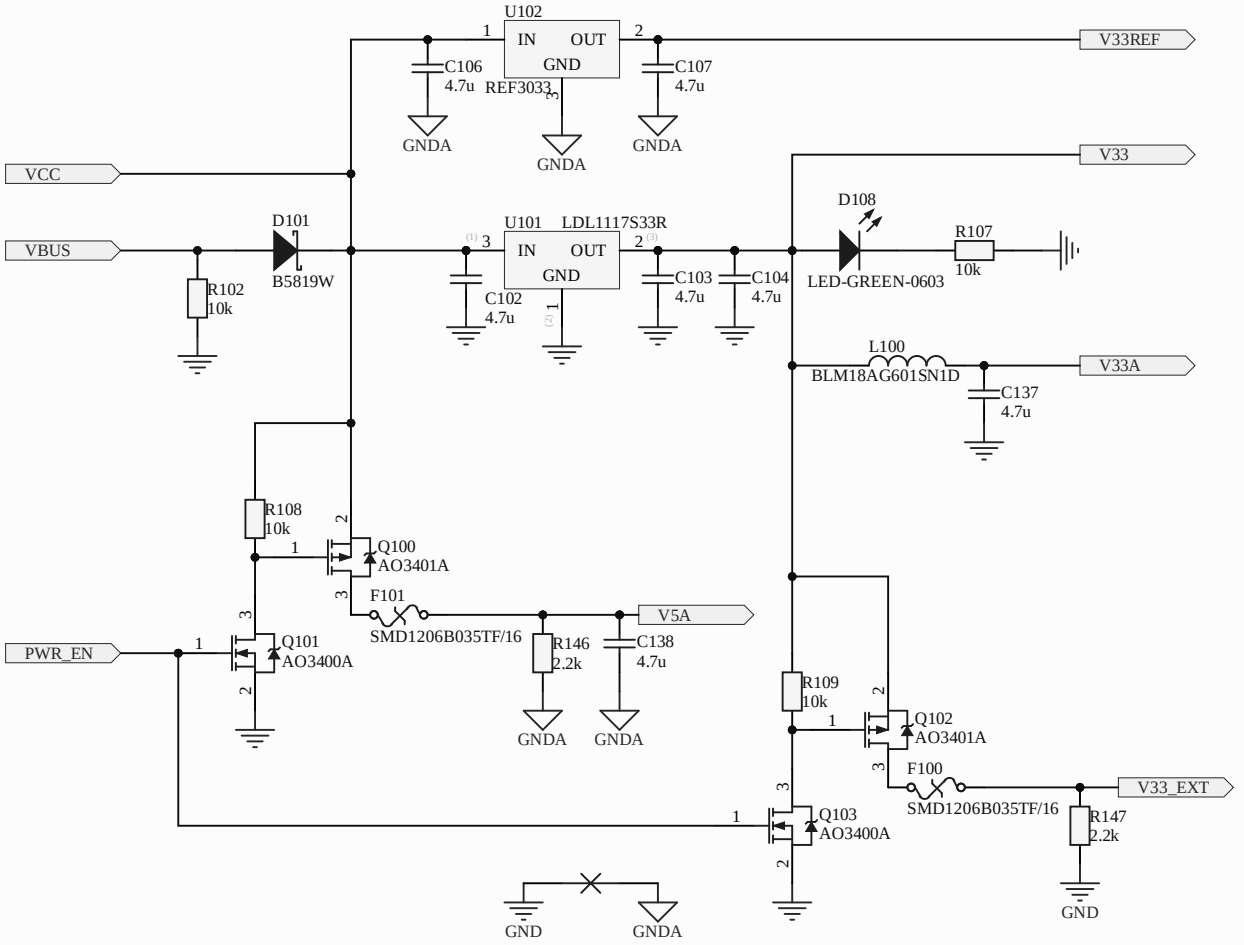
See pullups replacement file for actual values




U100
CAN_TX
CAN_TX
CAN_RX
CAN_RX
GND
VCC
1
2
3
4
TXD
GND
VCC
RXD
S
CANH
CANL
Vio
8
7
6
5
GND
V33
TJA1051T/3/1J
C105
100n
C100
100n
GND
GND
R100
120
CANH
CANL
D100
PESD1CAN
GND


M100
OUT_IGN1 W10
OUT_IGN2 W9
OUT_IGN3 W8
OUT_IGN4 W7
OUT_IGN5 W6
OUT_IGN6 W5
OUT_IGN7 W4
OUT_IGN8 W3
IGN1 (PC13)
IGN2 (PE5)
IGN3 (PE4)
IGN4 (PE3)
IGN5 (PE2)
IGN6 (PB8)
IGN7 (PB9)
IGN8 (PE6)
SPI2_SCK / CAN2_TX (PB13) E7 SPI2_SCK
SPI2_MISO (PB14) E8 SPI2_MISO
SPI2_MOSI (PB15) E9 SPI2_MOSI
SPI2_CS / CAN2_RX (PB12) E6 SPI2_CS
SPI3_SCK (PC10) N10 SPI3_SCK
SPI3_MISO (PC11) N11 SPI3_MISO
SPI3_MOSI (PC12) N12 SPI3_MOSI
SPI3_CS (PA15) N9 SPI3_CS
OUT_INJ1 E17
OUT_INJ2 E16
OUT_INJ3 E15
OUT_INJ4 E14
OUT_INJ5 E13
OUT_INJ6 E12
OUT_INJ7 E11
OUT_INJ8 E10
OUT_INJ1 (PD3)
OUT_INJ2 (PA9)
OUT_INJ3 (PD11)
OUT_INJ4 (PD10)
OUT_INJ5 (PD2)
OUT_INJ6 (PA8)
OUT_INJ7 (PD15)
OUT_INJ8 (PD12)
CANH W11 CANH
CANL W12 CANL
I2C_SDA (PB11) E4 I2C_SDA
I2C_SCL (PB10) E3 I2C_SCL
OUT_PWM1 E18
OUT_PWM2 E19
OUT_PWM3 E20
OUT_PWM4 E21
OUT_PWM5 E22
OUT_PWM6 E23
OUT_PWM1 (PD13)
OUT_PWM2 (PC6)
OUT_PWM3 (PC7)
OUT_PWM4 (PC8)
OUT_PWM5 (PC9)
OUT_PWM6 (PD14)
UART8_RX (PE0) N18 UART8_RX
UART8_TX (PE1) N19 UART8_TX
F40x does not have UART8
UART2_RX (PD6) N14 UART2_RX
UART2_TX (PD5) N13 UART2_TX
IN_CRANK S14
IN_CAM S16
IN_VSS S17
IN_CRANK (PB1)
IN_CAM (PA6)
IN_VSS (PE11)
USBP (PA12) N3 USB_P
USBM (PA11) N2 USB_N
USBID (PA10) N4 USBID
VBUS N1 VBUS
IN_TPS S18
IN_PPS S19
IN_TPS (PA4)
IN_PPS (PA3)
IN_IAT S20
IN_CLT S21
IN_IAT (PC3)
IN_CLT (PC2)
SWCLK (PA14) N6 SWCLK
SWDIO (PA13) N5 SWDIO
SWO (PB3) N8 SWO
IN_O2S S11
IN_O2S2 S10
IN_O2S / CAN_WAKEUP (PA0)
IN_O2S2 (PA1)
nReset N7 nRST
BOOT0 N16 BOOT0
IN_MAP1 S13
IN_MAP2 S12
IN_MAP1 (PC0)
IN_MAP2 (PC1)
LED_GREEN N14a LED_GREEN
LED_YELLOW N14b LED_YELLOW
IN_AUX1 S9
IN_AUX2 S8
IN_AUX3 S7
IN_AUX4 S6
IN_AUX1 (PB0)
IN_AUX2 (PC4/PE9)
IN_AUX3 (PC5)
IN_AUX4 (PA7)
IN_KNOCK S15
IN_KNOCK (PA2)
IN_D1 S4
IN_D2 S3
IN_D3 S2
IN_D4 S1
IN_D1 (PE12)
IN_D2 (PE13)
IN_D3 (PE14)
IN_D4 (PE15)
V33_SWITCHABLE N15 V33_EXT
Connected V33 N21 V33
V33 N23
V33 E1
VREF1 S22
VREF2 S5
V33REF W13
V33_REF
Connected V5A_SWITCHABLE W2 V5A
V5A_SWITCHABLE W14
V5A_SWITCHABLE W1
Connected GNDA E2
GNDA
PWR_EN N20
OUT_PWR_EN (PE10)
VIGN E5
VBAT N17
VCC N22
G
IN_VIGN (PA5)
VBAT
VCC
GND
Module:mega_mcu100/0.2
U_ign8
ign8.SchDoc
OUT_IGN1 IGN1
OUT_IGN2 IGN2
OUT_IGN3 IGN3
OUT_IGN4 IGN4
OUT_IGN5 IGN5
OUT_IGN6 IGN6
OUT_IGN7 IGN7
OUT_IGN8 IGN8
U_mcu
mcu.SchDoc
IGN1 IGN2 IGN3 IGN4 IGN5 IGN6 IGN7 IGN8
OUT_INJ1 OUT_INJ2 OUT_INJ3 OUT_INJ4 OUT_INJ5 OUT_INJ6 OUT_INJ7 OUT_INJ8
OUT_PWM1 OUT_PWM2 OUT_PWM3 OUT_PWM4 OUT_PWM5 OUT_PWM6
CAN_RX CAN_TX
SPI1_SCK SPI1_MISO SPI1_MOSI SPI1_CS1 SPI1_CS2
nReset
BOOT0
SPI2_SCK SPI2_MISO SPI2_MOSI SPI2_CS
SPI3_SCK SPI3_MISO SPI3_MOSI SPI3_CS
USBP USBM USBID VBUS
I2C_SDA I2C_SCL
UART8_RX UART8_TX
UART2_RX UART2_TX
SWCLK SWDIO SWO
LED_YELLOW LED_GREEN
PWR_EN
VIGN VBAT
IN_CRANK IN_CAM IN_VSS IN_TPS IN_PPS IN_IAT IN_CLT IN_O2S IN_O2S2 IN_MAP1 IN_MAP2 IN_AUX1 IN_AUX2 IN_AUX3 IN_AUX4 IN_KNOCK VREF1 VREF2
WAKEUP
U_can
can.SchDoc
CAN_RX CAN_TX CANL CANH
U_sd
sd.SchDoc
SD_SCK SD_MISO SD_MOSI SD_CS V33 V33_PULL V33 V33_EXT
U_sensors
sensors.SchDoc
ACC_SCK ACC_MISO ACC_MOSI ACC_CS V33 V33
BARO_SCL BARO_SDA
S100
nRST
C147
100n
S101 V33
I2C_SCL I2C_SDA
VCC
R142
10k
VBUS
VCC
PWR_EN
R148
0
R141
10k
U_power
power.SchDoc
VBUS VCC PWR_EN
V5A V33 V33A V33_EXT V33REF
P104 P106 P105
U_input
input.SchDoc
IN_CRANK IN_CAM IN_VSS
OUT_CRANK OUT_CAM OUT_VSS
IN_TPS IN_PPS
OUT_TPS OUT_PPS
IN_IAT IN_CLT
OUT_IAT OUT_CLT
IN_O2S IN_O2S2
OUT_O2S OUT_O2S2
IN_MAP1 IN_MAP2
OUT_MAP1 OUT_MAP2
IN_AUX1 IN_AUX2 IN_AUX3 IN_AUX4
OUT_AUX1 OUT_AUX2 OUT_AUX3 OUT_AUX4
IN_KNOCK
OUT_KNOCK
VREF1 VREF2
U_dinput4
dinput4.SchDoc
IN_D1 IN_D2 IN_D3 IN_D4
OUT_D1 OUT_D2 OUT_D3 OUT_D4
V33
R144
4.7k
U110
CAN_RX
2
4
5
3
SN74LVC1G06DCKR
WAKEUP


More accurate threshold for "raw" connection of VR sensors
R195
5.1k
R196
4.7k
GNDA
OUT1
OUT2
U195
1
2
3
IN1 (CRNK)
V5A
4
Vdd
C195
100n
GNDA
IN3
5
6
7
14
13
12
IN2
Vss
11
GNDA
10
9
8
IN4
MCP6004
R197
OUT3
OUT4
GNDA
4.7k



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Size A4	Number	Revision
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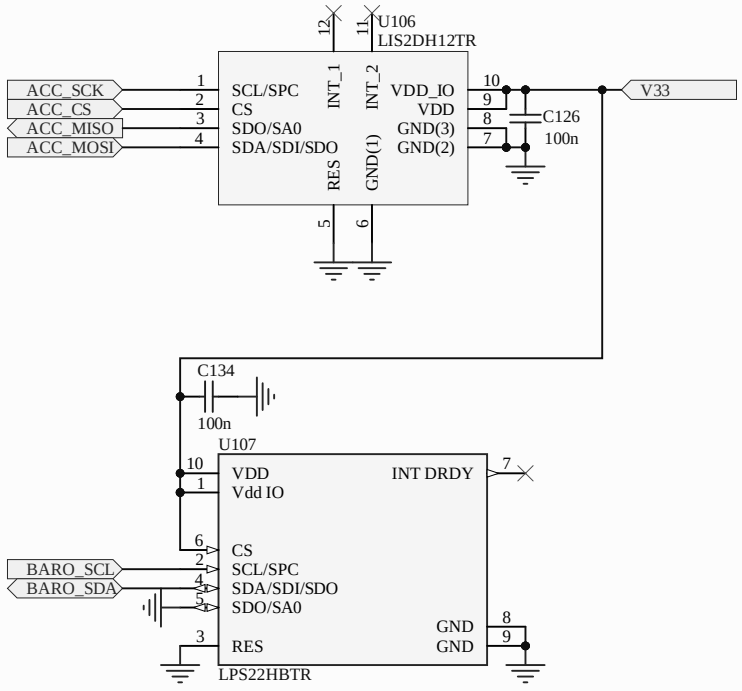
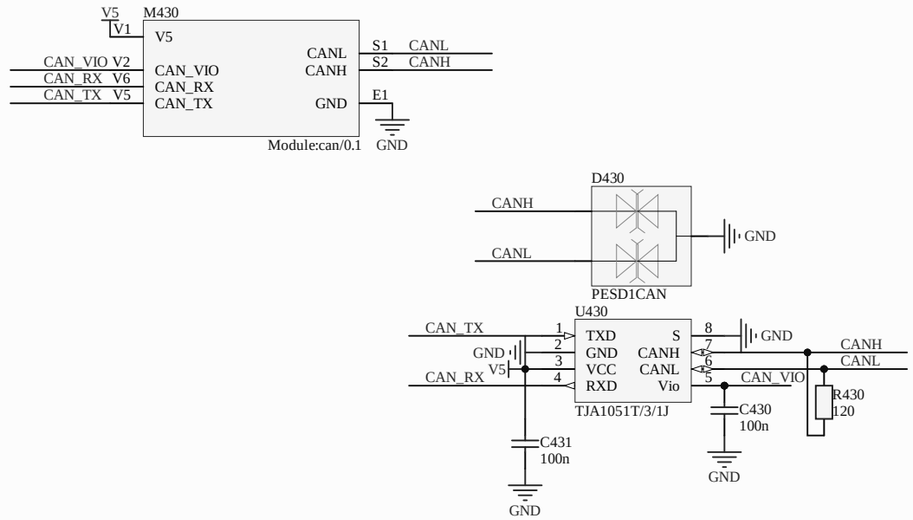
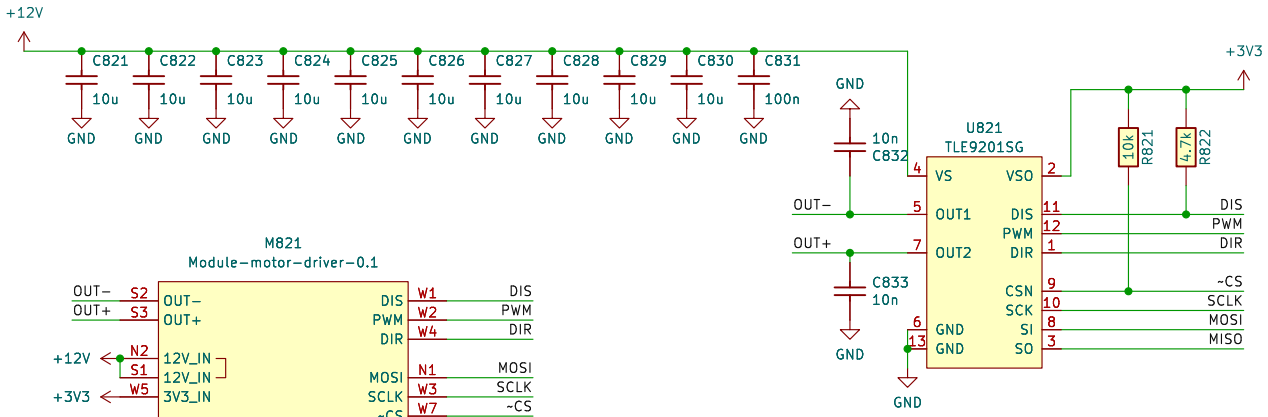

U106
LIS2DH12TR
ACC_SCK
ACC_CS
ACC_MISO
ACC_MOSI
SCL/SPC
CS
SDO/SA0
SDA/SDI/SDO
INT_1
INT_2
RES
GND(1)
VDD_IO
VDD
GND(3)
GND(2)
C126
100n
V33
C134
100n
U107
VDD
Vdd IO
INT DRDY
CS
SCL/SPC
SDA/SDI/SDO
SDO/SA0
RES
GND
GND
BARO_SCL
BARO_SDA
LPS22HBTR



Title: ***Hellen-One CAN***			***rusEFI EMS***
Size: A3	Author: andreika	Revision: 0.1	***Hellen-One Platform***
Date: 02.03.2021		Sheet 1 of 1	*https://rusefi.com/s/hellen72*



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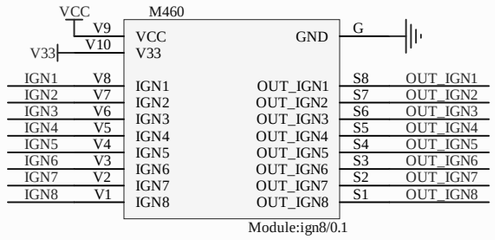
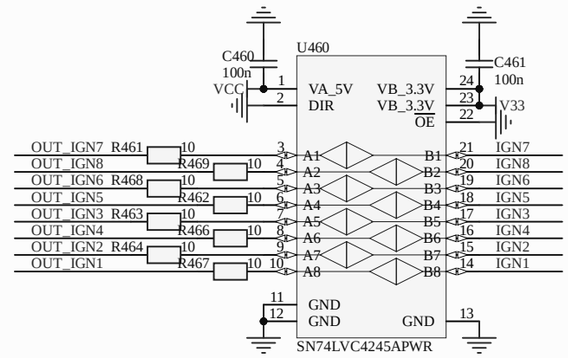
Title: Motor−driver module		
Size: A4	Date: 2024−03−07	**Rev: 0.1**
KiCad E.D.A. 8.0.3		Id: 1/1





Title: ***Hellen-One IGN8***			***rusEFI EMS*** ***Hellen-One Platform*** *https://rusefi.com/s/hellen1/*
Size: A3	Author: andreika	Revision: 0.2	
Date: 18/03/2022		Sheet 1 of 1	